

Computer Science & Information Technology

C Programming

DPP: 2

Control Flow Statement

Q1 #include<stdio.h>

```
int main(){
    int x = 0, y = 0, a;
    a = x && ++y;
    printf("%d %d", x, y);
    return 0;
}
```

what is the output of above program?

- (A) 11 (B) 10
(C) 01 (D) 00

Q2 #include<stdio.h>

```
int main()
{
    int x=3, y=4, z=4;
    printf("%d", (z>=y>=x?100:200));
    return 0 ;
}
```

what is the output of above program?

- (A) 100 (B) 200
(C) 0 (D) 1

Q3 #include <stdio.h>

```
int main(){ int a = 80;
switch(-12%45+36/9/2*16+60)
{
    case 80: a = a+10;
    case 5: a++;
    default : a = a>>2;
}
printf ("%d", a);
}
```

what is the output of above program?

- (A) 20 (B) 21
(C) 22 (D) 23

Q4 Question

int main()

```
{
    int a =50;
    switch(a)
    {
        default: a=45;
        case 49: a++;
        case 50: a--;
        case 51: a =a+1;
    }
```

```
printf("%d",a);
}
```

what is the output of the above program?

- (A) 51 (B) 45
(C) 50 (D) Error

Q5 Consider the following program

```
#include<stdio.h>
int main()
{
    int i= -1;
    for (; ++i; i++)
    {
        printf("I am a good Student");
    }
    return 0 ;
}
```

what is the output of the above program?

- (A) 0 (B) 1
(C) Infinite (D) 2

Q6 consider below c program.

```
#include <stdio.h>
int main()
{
    int i=2+4%6+9/10;
    while (i<10)
    {
```


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```
printf("I am good student");
i++;
}
return 0 ;
}
```

A

The number of times printf statement executed is _____

Q7 Consider the following C program

```
#include <stdio.h>
int main(){
float sum = 0.0, j = 1.0, i = 2.0;
while (i/j > 0.0625){
    j= j+j;
    sum = sum + i/j;
    printf("%f\n", sum);
}
return 0;
}
```

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what is the output of the above program?

Q8 #include <stdio.h>

```
int main() {
    int i,j;
    for(i=1;i<=3;i++){
        for(j=1;j<=20;j++){
            printf("I am a good student");
            if(i==2) break;
        }
    }
    return 0 ;
}
```

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The number of times printf will be executed is_____.

Q9 Consider the c program given below.

```
#include<stdio.h>
void main () {
    int sum =0;
    for (int j =1; j<=20; j++)
    for (int i = 1; i <= 10; i++) {
        if (i == j)
            continue;
        sum++;
    }
}
```

```
printf("%d\n", sum);
}
```

The output of the program is _____

(A) 200

(B) 190

(C) 180

(D) 20

Q10 Consider the function func shown below:

```
#include <stdio.h>
int main() {
    int num=128;
    int count = 0;
    while (num/2) {
        count++;
        num>>= 1;
    }
    printf ("%d", count);
}
```

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The value printed is _____.

Q11 Consider the function func shown below:

```
#include <stdio.h>
int main()
{
    int a=7, b=8;
    while(++b & a-- )
    {
        printf("Hello!");
    }
    return 0;
}
```

2

The number of times the printf() executed is _____.

Q12 Consider the function func shown below:

```
int main() {
    int a=1, b=2;
    do
    {
        while(b++)
        {
            b=b-a;
            a=a+b;
        }
    }
    while(a++<2);
}
```


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```
printf("%d\t%d", a, b);  
return 0;  
}
```

The output is_____

(A) 3 0

(B) 4 2

(C) 3 2

(D) 4 1



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Answer Key

Q1 (D)

Q2 (B)

Q3 (C)

Q4 (C)

Q5 (A)

Q6 4

Q7 1.9375

Q8 41

Q9 (B)

Q10 7

Q11 7

Q12 (D)



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Hints & Solutions

Q1 Text Solution:

$x' = 0, y = 0, a$
 $a = 0 \ \&\& \ ++y$ will not execute logical $\rightarrow$ short circuit code.
 Logic short

Q2 Text Solution:

$x = 3, y = 4, z = 4$
 conditional operator $\geq$ left associative operator
 $(r \geq 4 > = 3)$
 $1 \geq 3$ False

Q3 Text Solution:

$a = 80$
 $-12 \% 45 + 36 / 9 / 2 * 16 + 60$
 $= -12 + 4/2 * 16 + 60$
 $= -12 + 32 + 60$
 $= 20 + 60 = 80$
 Case 80 : $a = 90$
 Case 5; $a = 91$
 $a = 91 \gg 2$
 $\lfloor 91/2^2 \rfloor = \lfloor 91/4 \rfloor$
 $= 22$

Q4 Text Solution:

`int a = 50`
`switch(a)`
`switch(50)`
 case 50 will match
 hence option c is correct.

Q5 Text Solution:

for loop
 $i = -1$
 for ($;; ++i; i++$)
 No initialization
 $++i = i$
 $i = -1 \rightarrow 0$
 condition is false
 No – times – 0 time

Q6 Text Solution:

$i = 2 + 4\%6 + /10$
 $= 2 + 4 + 0 = 6$

 $i = 6$
 $6 < 10 \quad \dots(1)$
 $7 < 10 \quad \dots(2)$
 $8 < 10 \quad \dots(3)$
 $9 < 10 \quad \dots(4)$
 $11 < 10 \quad \times$

Number of times print execute = 4

Q7 Text Solution:
 $i/j > 0.0625$
 $(1) 2/1 > 0.0625$
 $j = 2$
 $\text{sum} = 0 + 2/2 = 1$
 $(2) 2/2 > 0.0625$
 $j = 4$
 $\text{sum} 2/4 + 1 = 1.5$
 $(3) 2/4 > 0.0625$
 $j = 8$
 $\text{sum} = 1.5 + 0.25 = 1.75$
 $(4) 2/8 > 0.0625$
 $j = 16$
 $\text{sum} = 1.75 + 0.125 = 1.875$
 $(5) 2/16 > 0.0625$
 0.125
 $j = 32$
 $\text{sum} = 1.875$
 $\underline{0.0625}$
 1.9375
 $(6) 2/32 > 0.0625$ false

Final value of sum = 1.9375

Q8 Text Solution:

$i = 1$ $\left\{ \begin{array}{l} j = 1 \\ j = 20 \end{array} \right\}$ 20 times

$i = 3$ — $j = 1$ print

$i = 3$ $\left\{ \begin{array}{l} j = 1 \\ j = 20 \end{array} \right\}$ 20 times

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times printf execute

Q9 Text Solution:

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for loop

```

    j = 1
    j = 2
i = 1
    :
    j = 10
    j = 1
    j = 2
i = 2
    :
    j = 10
i = 3
:
:
i = 20
200 times 20 × 10 = 200
10 case i = 1, j = 1
i = 2, j = 2
i = 3, j = 3
200 - 10 = 190

```

Q10 Text Solution:

Num = 128

Count = 0

While (Condition)	count	num
128/2 = 64	1	64
64/2 = 32	2	32
32/2 = 16	3	16
16/2 = 8	4	8
8/2 = 4	5	4
4/2 = 2	6	2

2/2 = 1

7

1

1/2 = 0 condition is false

Q11 Text Solution:

```

a = 7,      b = 8
a = 6,      b = 9 ....(1)
00111
01001
00001 Non
00110
01010 Non -
01011 .....(3)
00101 Non
a = 3,      b = 12
01100.....(4)
00100 Non
a = 2,      b = 13
01101.....(5)
00011 Non
a = 1,      b = 14
01110.....(6)
00010 Non zero
a = 0,      b = 15
01111.....(7)
00001 Non zero
a = 0,      b = 16

```

Q12 Text Solution:

a = 1, b = 2 3 2 3

while (2)

b = 3 - 1 = 2

a = 1 + 2 = 3

While (2)

b = 3 - 3 = 0

a = 3 + 0 = 3

while (0)

after loop

3 < 2 false

come at of loop

post increment

a = 4

b = 1

Correction option 'd'.



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